· $\langle n' | n \rangle = \delta_{n'n}$ （正交性）

· $\sum_{n=0}^{\infty} |n\rangle\langle n| = \mathbb{1}$ （完备性）

总之：

$$\hat{H} = \left(\hat{a}^{+}\hat{a} + \frac{1}{2}\right)\hbar\omega$$

$$\begin{cases} \hat{H}|n\rangle = E_n |n\rangle, \\ |n\rangle = (\hat{a}^{+})^n/\sqrt{n!}\,|0\rangle \\ E_n = \left(n + \frac{1}{2}\right)\hbar\omega, \quad n = 0, 1, 2, \cdots \end{cases}$$

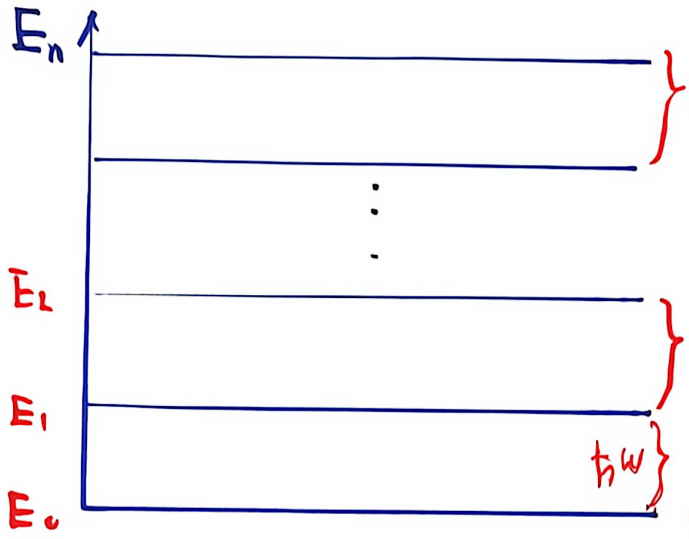
·能谱特征：

基态：$n = 0 \Rightarrow |0\rangle$

基态能：$E_0 = \frac{1}{2}\hbar\omega$.

激发态：$n > 0 \Rightarrow |n\rangle$

激发能：$E_n - E_0 = n\hbar\omega$



等间隔的

(其激发能为即 $n\hbar\omega$)

粒子数表象：将 $|n\rangle$ 态看成有 n 个准粒子被激发的态，每个准粒子的能量都等于 $\hbar\omega$。以 $\{|n\rangle\}$ 为基矢的表象称为粒子数表象.

$\hat{a}^\dagger$ ：产生算符
$\hat{a}$ ：湮灭算符
$\hat{N}$ ：粒子数算符

(晶格振动系统，自由电磁场，全同自旋玻色子系统等)

引入 $\hat{a}_k$, $\hat{a}_k^\dagger$ $(k = k_1, k_2, k_3 \cdots)$

↳ 量子数

$$\begin{cases} [\hat{a}_k, \hat{a}_{k'}^\dagger] = \delta_{kk'} \\ [\hat{a}_k, \hat{a}_{k'}] = [\hat{a}_k^\dagger, \hat{a}_{k'}^\dagger] = 0 \end{cases}$$

$$\hat{H}_0 = \sum_k \hbar\omega_k \hat{a}_k^\dagger \hat{a}_k + \text{const.}$$

引入 k态粒子数算符： $\hat{N}_k = \hat{a}_k^\dagger \hat{a}_k$

则. $\hat{H}_0 = \sum_k \hbar\omega_k \hat{N}_k = \sum_k \hat{h}_k$, $\hat{h}_k = \hbar\omega_k \hat{N}_k$

$$\begin{cases} \hat{N}_k |n_k\rangle = n_k |n_k\rangle \\ n_k = 0, 1, 2, \cdots \\ |n_k\rangle = \frac{1}{\sqrt{n_k!}} (\hat{a}_k^\dagger)^{n_k} |0\rangle \end{cases}$$

$[\hat{N}_k, \hat{N}_{k'}] = 0$, $[\hat{H}, \hat{N}_k] = 0$

共同本征态：

- $|n_{k_1}\ n_{k_2}\cdots n_{k_n}\cdots\rangle$

$$=\prod_k |n_k\rangle = \prod_k \frac{1}{\sqrt{n_k!}}(\hat{a}_k^+)^{n_k}|0\rangle$$

$$\begin{cases} \hat{N}_{k_n}|n_{k_1}, n_{k_2}\cdots n_{k_n}\cdots\rangle = n_{k_n}|n_{k_1}, n_{k_2}\cdots n_{k_n}\cdots\rangle \\ \hat{H}_0|n_{k_1}, n_{k_2}\cdots n_{k_n}\cdots\rangle = \sum_n \hbar\omega_{k_n} n_{k_n} \end{cases}$$

$$\begin{cases} \langle n_{k_1}\ n_{k_2}\cdots n_{k_n}\cdots | n_{k_1'}\ n_{k_2'}\cdots n_{k_n'}\cdots\rangle \\ \quad = \delta_{k_1k_1'}\delta_{k_2k_2'}\cdots\delta_{k_nk_n'}\cdots \quad \text{正交性} \\ \\ \sum_{\{n\}} |n_{k_1}\ n_{k_2}, n_{k_n}\cdots\rangle\langle n_{k_1}\ n_{k_2}\cdots n_{k_n}\cdots| = \mathbb{1} \quad \text{完备性} \end{cases}$$

- $|\psi\rangle = \mathbb{1}|\psi\rangle = \sum_{\{n\}} |n_{k_1}, n_{k_2}\cdots n_{k_n}\cdots\rangle \langle n_{k_1}\ n_{k_2}\cdots n_{k_n}\cdots|\psi\rangle$.

- 粒子表象

$\hat{a}_k$：消灭算符，$\hat{a}_k^+$：产生算符

$|n\rangle$ 的坐标表象：$\langle x|n\rangle \equiv \psi_n(x)$

定态波函数

基态波函数 $\psi_0(x) = \langle x|0\rangle$，$\left(E_0 = \frac{\hbar\omega}{2}\right)$

由 $\hat{a}|0\rangle = 0|零\rangle$　　$\hat{x}|x\rangle = x|x\rangle$

$\Rightarrow \langle x|\hat{a}|0\rangle = 0$，　$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i\hat{p}}{m\omega}\right)$

$\Rightarrow \langle x|\left[\sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i\hat{p}}{m\omega}\right)\right]|0\rangle = 0$，

$\Rightarrow \int_{-\infty}^{\infty} dx' \langle x|\left[\sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i\hat{p}}{m\omega}\right)\right]|x'\rangle\langle x'|0\rangle = 0$.

第一项：$\sqrt{\frac{m\omega}{2\hbar}} \int_{-\infty}^{\infty} dx' \langle x|x'\rangle\langle x'|0\rangle x'$

$= \sqrt{\frac{m\omega}{2\hbar}} \int_{-\infty}^{\infty} dx' \delta(x-x')\langle x'|0\rangle x'$

$= \sqrt{\frac{m\omega}{2\hbar}} \langle x|0\rangle x = \sqrt{\frac{m\omega}{2\hbar}} \psi_0(x) x$.

$$\text{右}=\sqrt{\frac{m\omega}{2\hbar}}\frac{i}{m\omega}\int_{-\infty}^{\infty}dx'\langle x|\hat{p}|x'\rangle\langle x'|0\rangle$$

但 $\int_{-\infty}^{\infty}dx'\ p_{xx'}\langle x'|0\rangle$

$$=\int_{-\infty}^{\infty}dx'\left(-i\hbar\frac{\partial}{\partial x}\delta(x-x')\right)\langle x'|0\rangle$$

$$=\left(-i\hbar\frac{\partial}{\partial x}\right)\int_{-\infty}^{\infty}dx'\ \delta(x-x')\langle x'|0\rangle$$

$$=\left(-i\hbar\frac{\partial}{\partial x}\right)\langle x|0\rangle=\left(-i\hbar\frac{\partial}{\partial x}\psi_0(x)\right)$$

$$\Rightarrow\sqrt{\frac{m\omega}{2\hbar}}\left[x+\frac{\hbar}{m\omega}\frac{\partial}{\partial x}\right]\psi_0(x)=0,$$

$$\Rightarrow\left(x+x_0^2\frac{\partial}{\partial x}\right)\psi_0(x)=0,\quad\left(x_0\equiv\sqrt{\frac{\hbar}{m\omega}}\right)$$

$$\Rightarrow\psi_0(x)=A\exp\left[-\frac{1}{2}\left(\frac{x}{x_0}\right)^2\right]$$

（A为待定常数）

- $\langle 0|0\rangle = 1$,

$$\int_{-\infty}^{\infty} dx \, \langle 0|x\rangle\langle x|0\rangle = 1 \quad ,$$

$$\Rightarrow \int_{-\infty}^{\infty} dx \, \psi_0^*(x)\,\psi_0(x) = 1$$ 규격화 조건.

$$\Rightarrow \boxed{A = \frac{1}{\pi^{\frac{1}{4}}\sqrt{x_0}}}$$

- 첫 번째 들뜬 상태 파동함수 $\psi_1(x) = \langle x|1\rangle$,

$$\left(E_1 = \frac{3}{2}\hbar\omega\right)$$

$$\psi_1(x) = \langle x|1\rangle = \langle x|\hat{a}^+|0\rangle$$

$$= \langle x|\sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i\hat{p}}{m\omega}\right)|0\rangle$$

$$= \int_{-\infty}^{\infty} dx' \, \langle x|\sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i\hat{p}}{m\omega}\right)|x'\rangle\langle x'|0\rangle$$

$$= \frac{1}{\sqrt{2}x_0}\left(x - x_0^2\frac{\partial}{\partial x}\right)\psi_0(x)$$

· 第二激发态波函数 $\psi_2(x) = \langle x|2\rangle$,

$E_2 = \frac{5}{2}\hbar\omega$.

$$\psi_2(x) = \langle x|2\rangle = \langle x| \frac{1}{\sqrt{2!}}(\hat{a}^+)^2|0\rangle$$

$$= \frac{1}{\sqrt{2!}}\left(\frac{1}{\sqrt{2}x_0}\right)^2\left(x - x_0^2\frac{\partial}{\partial x}\right)^2\psi_0(x),$$

$$\vdots$$

$$\psi_n(x) = \left(\frac{1}{\pi^{\frac{1}{4}}\sqrt{2^n n!}}\right)\left(\frac{1}{x_0^{n+\frac{1}{2}}}\right)\left(x - x_0^2\frac{\partial}{\partial x}\right)^n \exp\left[-\frac{1}{2}\left(\frac{x}{x_0}\right)^2\right]$$

$$= \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}}\frac{1}{\sqrt{2^n n!}} H_n(\xi)\, e^{-\frac{1}{2}\xi^2}$$

其中 $\xi \equiv \frac{x}{x_0}$, $H_n(\xi) = e^{\frac{\xi^2}{2}}\left(\xi - \frac{d}{d\xi}\right)^n e^{-\frac{\xi^2}{2}}$

厄米多项式.

- 数学公式

$$\sum_{n=0}^{\infty} t^n \frac{H_n(\xi)}{n!} = \exp\left[2\xi t - t^2\right]$$

其中 $H_n(\xi)$ 为厄米多项式

- $$\left[\hat{a}, \hat{f}(\hat{a}, \hat{a}^+)\right] = \frac{\partial}{\partial \hat{a}^+} \hat{f}(\hat{a}, \hat{a}^+)$$

- $$\left[\hat{a}^+, \hat{f}(\hat{a}, \hat{a}^+)\right] = -\frac{\partial}{\partial \hat{a}} \hat{f}(\hat{a}, \hat{a}^+)$$

其中 $\hat{f}(\hat{a}, \hat{a}^+)$ 为可展开成 $\hat{a}$ 与 $\hat{a}^+$ 的幂级数的算符函数.

§1.5 谐振子相干态 $|\alpha\rangle$

Coherent State

$\hat{H} = \left(\hat{a}^+\hat{a} + \frac{1}{2}\right)\hbar\omega$

1、定义式

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$$

- 谐振子相干态定义为谐振子系统消灭算符 $\hat{a}$ 的本征态.

$\hat{a}$:

$$\begin{cases} [\hat{a}, \hat{a}^+] = 1, \quad \hat{a}^+\hat{a}|n\rangle = n|n\rangle, \\ \hat{a}|n\rangle = \sqrt{n}|n-1\rangle \ (n>1), \quad \hat{a}|0\rangle = 0 \end{cases}$$

($\hat{a}^+\hat{a} = \hat{N}$)

※ $\hat{a}$ 不是厄米算符，所以本征值 α 一般为复数!

2. 相干态 $|\alpha\rangle$ 的具体表达式

$$\hat{a}\,|\alpha\rangle = \alpha\,|\alpha\rangle ,$$

($\langle\alpha|\alpha\rangle = 1$)

$$|\alpha\rangle = \sum_{n=0}^{\infty} |n\rangle\langle n|\alpha\rangle .$$

由 $\quad |n\rangle = \dfrac{(\hat{a}^{+})^{n}}{\sqrt{n!}}\,|0\rangle$

$$\Rightarrow \quad \langle n| = \langle 0|\,\frac{(\hat{a})^{n}}{\sqrt{n!}} .$$

$$\Rightarrow \quad \langle n|\alpha\rangle = \langle 0|\,\frac{(\hat{a})^{n}}{\sqrt{n!}}\,|\alpha\rangle$$

$$= \frac{\alpha^{n}}{\sqrt{n!}}\langle 0|\alpha\rangle .$$

$$\Rightarrow \quad |\alpha\rangle = \langle 0|\alpha\rangle \sum_{n=0}^{\infty} \frac{\alpha^{n}}{\sqrt{n!}}\,|n\rangle$$

确定

[由归一化条件 $\langle\alpha|\alpha\rangle = 1$]

$$\langle \alpha | \alpha \rangle = \left(\langle 0|\alpha \rangle^* \sum_{m=0}^{\infty} \langle m | \frac{(\alpha^*)^m}{\sqrt{m!}} \right) \left(\langle 0|\alpha \rangle \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle \right)$$

$$= |\langle 0|\alpha \rangle|^2 \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \frac{(\alpha^*)^m}{\sqrt{m!}} \frac{(\alpha)^n}{\sqrt{n!}} \underbrace{\langle m|n \rangle}_{\delta_{mn}}$$

$$= |\langle 0|\alpha \rangle|^2 \sum_{n=0}^{\infty} \frac{(\alpha^*)^n (\alpha)^n}{n!}$$

$$= |\langle 0|\alpha \rangle|^2 \sum_{n=0}^{\infty} \frac{(|\alpha|^2)^n}{n!}$$

$$= |\langle 0|\alpha \rangle|^2 \exp\left[|\alpha|^2 \right] = \underline{1}$$

$$\Rightarrow \quad \langle 0|\alpha\rangle = e^{-|\alpha|^2/2} \qquad (\text{取为实数})$$

$$\Rightarrow \quad \boxed{|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle}$$

※ α 的定义域为整个复平面 (连续谱)

· $|\alpha\rangle$ 的另一种表达式

$$\boxed{|\alpha\rangle = \hat{D}(\alpha)|0\rangle}$$

$$\hat{D}(\alpha) \equiv e^{\alpha \hat{a}^\dagger - \alpha^* \hat{a}} \qquad (\text{位移算符})$$

$\hat{a}^\dagger$ 的本征态？

证明：

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$

另一种表达

$$= e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} \frac{(\hat{a}^+)^n}{\sqrt{n!}} |0\rangle$$

$$= e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{(\alpha \hat{a}^+)^n}{n!} |0\rangle$$

$$= e^{-|\alpha|^2/2} \; e^{\alpha \hat{a}^+} |0\rangle$$

另一种表达

$$= e^{-|\alpha|^2/2} \; e^{\alpha \hat{a}^+} \; e^{-\alpha^* \hat{a}} |0\rangle$$

$\because$ $e^{-\alpha^* \hat{a}} |0\rangle = \left(1 - \alpha^* \hat{a} + \frac{(\alpha^* \hat{a})^2}{2} + \cdots\right)|0\rangle$

$= (|0\rangle - 0 + 0 - \cdots) = |0\rangle$

Glauber 公式：

$$e^{\hat{A}+\hat{B}} = e^{\hat{A}} e^{\hat{B}} e^{-[\hat{A},\hat{B}]/2} = e^{\hat{B}} e^{\hat{A}} e^{[\hat{A},\hat{B}]/2},$$

if $[\hat{A},[\hat{A},\hat{B}]] = [\hat{B},[\hat{A},\hat{B}]] = 0$

取 $\hat{A} = \alpha \hat{a}^\dagger$, $\hat{B} = -\alpha^* \hat{a}$

则有 $[\hat{A}, \hat{B}] = |\alpha|^2$ (常数)

$\Rightarrow$ $[\hat{A}, [\hat{A}, \hat{B}]] = 0$,

$[\hat{B}, [\hat{A}, \hat{B}]] = 0$.

$\Rightarrow$ $e^{\alpha \hat{a}^\dagger} e^{-\alpha^* \hat{a}}$

$= e^{\alpha \hat{a}^\dagger - \alpha^* \hat{a}} e^{|\alpha|^2/2}$.

$\Rightarrow$ $|\alpha\rangle = e^{-|\alpha|^2/2} e^{\alpha \hat{a}^\dagger} e^{-\alpha^* \hat{a}} |0\rangle$

$= e^{-|\alpha|^2/2} e^{\alpha \hat{a}^\dagger - \alpha^* \hat{a}} e^{|\alpha|^2/2} |0\rangle$

$= e^{\alpha \hat{a}^\dagger - \alpha^* \hat{a}} |0\rangle$

$= \hat{D}(\alpha) |0\rangle$. 得证.

位移算符的性质

- $\hat{D}(\alpha=0) = \mathbb{1}$，$|\alpha\rangle = \hat{D}(\alpha)|0\rangle$ 中

令 $\alpha=0 \Rightarrow |\alpha=0\rangle = \hat{D}(\alpha=0)|0\rangle = \mathbb{1}|0\rangle = |0\rangle$

谐振子基态 $|0\rangle$ 是相干态！（本征值 $\alpha=0$）

- $\hat{D}^\dagger(\alpha) = \hat{D}(-\alpha) = \hat{D}^{-1}(\alpha)$，（幺正算符）
- $\hat{D}(\alpha)\ \hat{a}\ \hat{D}^\dagger(\alpha) = \hat{a} - \alpha$，
- $\hat{D}(\alpha)\ \hat{x}\ \hat{D}^\dagger(\alpha) = \hat{x} - \bar{x}_\alpha$，
- $\hat{D}(\alpha)\ \hat{p}\ \hat{D}^\dagger(\alpha) = \hat{p} - \bar{p}_\alpha$，

（$\bar{x}_\alpha \equiv \sqrt{\frac{2\hbar}{m\omega}}\,\mathrm{Re}[\alpha]$，$\bar{p}_\alpha \equiv \sqrt{2\hbar m\omega}\,\mathrm{Im}[\alpha]$）

3. 相干态的若干性质

(a) 相干态的坐标表象 $\langle x|\alpha\rangle \equiv \psi_\alpha(x)$

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle, \quad \text{左乘} \langle x|,$$

$$\Rightarrow \quad \langle x|\hat{a}|\alpha\rangle = \alpha\langle x|\alpha\rangle,$$

$$\Rightarrow \quad \langle x|\left[\sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x}+\frac{i\hat{p}}{m\omega}\right)\right]|\alpha\rangle = \alpha\langle x|\alpha\rangle.$$

左边为-2项: $\sqrt{\frac{m\omega}{2\hbar}}\langle x|\hat{x}|\alpha\rangle = \sqrt{\frac{m\omega}{2\hbar}}\, x\langle x|\alpha\rangle.$

左边为=2项: $\sqrt{\frac{m\omega}{2\hbar}}\frac{i}{m\omega}\langle x|\hat{p}|\alpha\rangle$

$$= \sqrt{\frac{m\omega}{2\hbar}}\frac{i}{m\omega}\int_{-\infty}^{\infty} dx' \langle x|\hat{p}|x'\rangle\langle x'|\alpha\rangle$$

$$= \sqrt{\frac{m\omega}{2\hbar}}\frac{i}{m\omega}\left(-i\hbar\frac{\partial}{\partial x}\right)\int_{-\infty}^{\infty} dx' \,\delta(x-x')\langle x'|\alpha\rangle$$

$$= \sqrt{\frac{m\omega}{2\hbar}}\frac{\hbar}{m\omega}\frac{\partial}{\partial x}\langle x|\alpha\rangle .$$

将 $\langle x|\alpha\rangle = \psi_\alpha(x)$ 代入上两式，得

$$\left[x - \sqrt{2}\,x_0\alpha + x_0^2\frac{\partial}{\partial x}\right]\psi_\alpha(x) = 0 \quad \left(x_0 = \sqrt{\frac{\hbar}{m\omega}}\right)$$

一般解：$\psi_\alpha(x) = A\exp\left[-\frac{(x-\sqrt{2}x_0\alpha)^2}{2x_0^2}\right]$.

确定 A：$\langle\alpha|\alpha\rangle = 1 \Rightarrow$

$$\int dx\,\langle\alpha|x\rangle\langle x|\alpha\rangle = 1 \Rightarrow$$

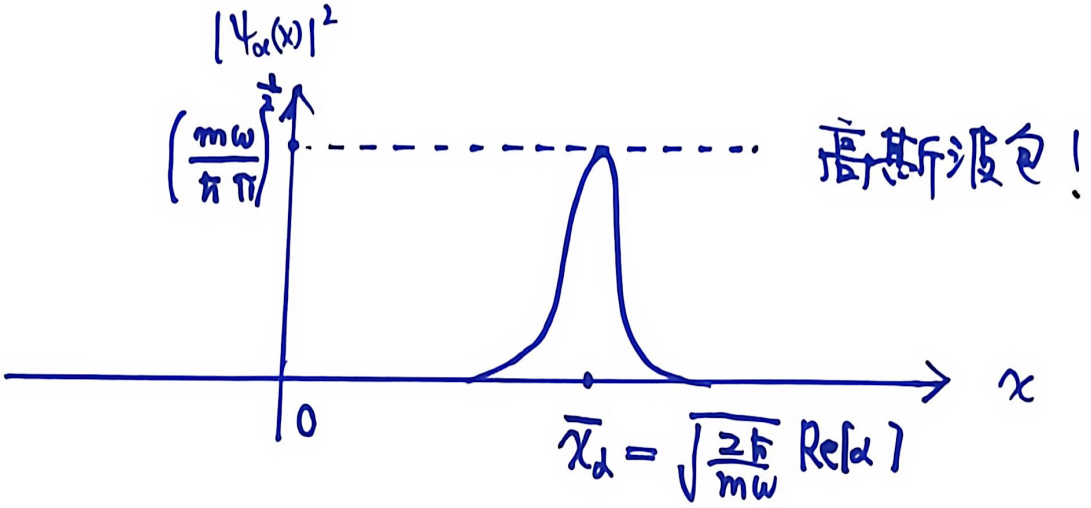
$$\int dx\,|\langle x|\alpha\rangle|^2 = 1 \Rightarrow$$

$$\int dx\,|\psi_\alpha(x)|^2 = 1 .$$

可取：$A = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}}\exp\left[-I_m\alpha^2\right]\exp\left[i\sqrt{2}\,Re\alpha\cdot I_m\alpha\right]$

$$\Rightarrow \psi_\alpha(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}}\exp\left[-\frac{1}{2}\left\{\frac{x-\bar{x}_\alpha}{x_0}\right\}^2 + i\frac{\bar{p}_\alpha x}{\hbar}\right].$$

· $|\psi_\alpha(x)|^2 = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{2}} \exp\left[-\left\{\frac{x-\bar{x}_\alpha}{x_0}\right\}^2\right]$



相干迭加！

(b) 相干态随时间的演化

设 $t=0$ 时刻谐振子处于相干态 $|\alpha\rangle$，

则 $t>0$ 时刻演化为 $|\alpha,t\rangle$ 态，

$$|\alpha,t\rangle = e^{-\frac{i}{\hbar}\hat{H}t}|\alpha\rangle .$$

$$\hat{H} = \left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)\hbar\omega .$$

$t=0$, $|\alpha\rangle$

· $|\alpha,t\rangle$ 是相干态吗？

$$|\alpha,t\rangle = e^{-\frac{i}{\hbar}\hat{H}t}|\alpha\rangle = e^{-\frac{i}{\hbar}(\hat{a}^\dagger\hat{a}+\frac{1}{2})\hbar\omega t}|\alpha\rangle$$

$$= e^{-i\omega t/2}\, e^{-i\hat{a}^\dagger\hat{a}\omega t}|\alpha\rangle$$

$$= e^{-i\omega t/2}\, e^{-i\hat{N}\omega t}\, e^{\frac{|\alpha|^2}{2}}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}|n\rangle$$

$$= e^{-i\omega t/2}\, e^{-|\alpha|^2/2}\sum_{n=0}^{\infty} e^{-i\hat{N}\omega t}|n\rangle\frac{\alpha^n}{\sqrt{n!}}$$

$$= e^{-i\omega t/2}\, e^{-|\alpha|^2/2}\sum_{n=0}^{\infty} e^{-in\omega t}\frac{\alpha^n}{\sqrt{n!}}|n\rangle$$

$$= e^{-i\omega t/2}\, e^{-|\alpha e^{-i\omega t}|^2/2}\sum_{n=0}^{\infty}\frac{(\alpha e^{-i\omega t})^n}{\sqrt{n!}}|n\rangle$$

与相干态 $|\alpha\rangle$ 的表达式

$$|\alpha\rangle = e^{-|\alpha|^2/2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}|n\rangle$$ 相比较.

$$\alpha \to \alpha e^{-i\omega t}$$

$$\Rightarrow \underline{|\alpha,t\rangle = e^{-i\omega t/2}\,|\alpha e^{-i\omega t}\rangle},$$

$\underline{\hat{a}|\alpha,t\rangle = \alpha e^{-i\omega t}|\alpha,t\rangle}$ 相干态！

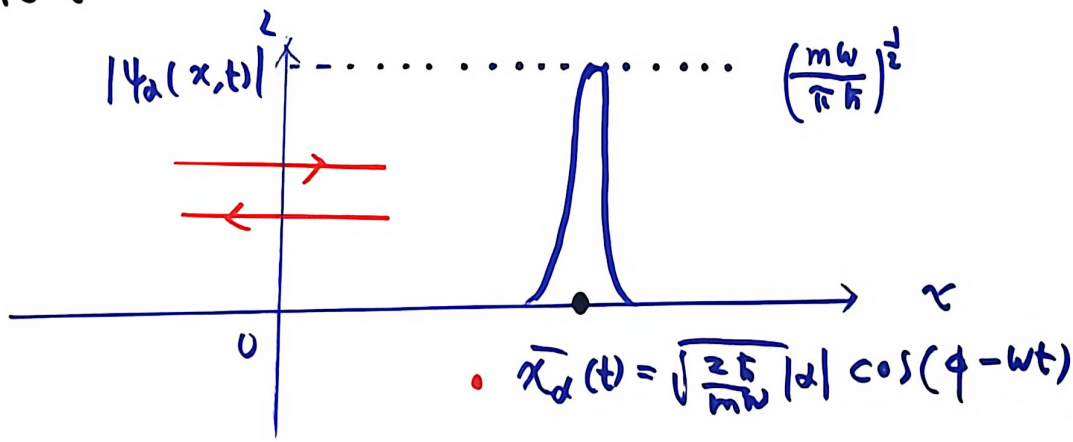
$$\cdot \quad |\langle x|\alpha,t\rangle|^2 \equiv |\psi_\alpha(x,t)|^2 .$$

得. $$|\psi_\alpha(x,t)|^2 = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{2}} \exp\left[-\left\{\frac{x-\bar{x}_\alpha(t)}{x_0}\right\}^2\right] ,$$

$$\bar{x}_\alpha(t) \equiv \sqrt{\frac{2\hbar}{m\omega}}\, \mathrm{Re}\left[\alpha\, e^{-i\omega t}\right] \quad \text{(波包中心)}$$

令 $\alpha = |\alpha| e^{i\phi}$.

$$\Rightarrow \quad \mathrm{Re}\left[\alpha\, e^{-i\omega t}\right] = |\alpha| \cos[\phi - \omega t]$$



波包不扩散，而波包中心严格时间作简谐振动！ (类似于经典谐振子)

(C) 概率分布 $|\langle n|\alpha,t\rangle|^2 \equiv P_\alpha(n,t) = ?$

(单色激光中的光子数)

- $P_\alpha(n,t) = e^{-|\alpha e^{i\omega t}|^2} \frac{|\alpha e^{i\omega t}|^{2n}}{n!}$

$$= e^{-|\alpha|^2} \frac{|\alpha|^{2n}}{n!} = P_\alpha(n) \text{ (与时间无关.)}$$

$$\bar{n} \equiv \langle \alpha,t|\ \hat{N}\ |\alpha,t\rangle = |\alpha e^{i\omega t}|^2$$

$$= |\alpha|^2$$

$$\Rightarrow \quad P_\alpha(n,t) = e^{-\bar{n}} \frac{\bar{n}^n}{n!}$$

(当 $n = |\alpha|^2$ 的整数部分时, P_α 取极大)

经典泊松分布:

事情平均发生 $\bar{n}$ 次的情况下.

实际发生 n 次的概率.

- 相对涨落: $\sqrt{\overline{(\Delta\hat{N})^2}}/\bar{n} = \frac{|\alpha|}{|\alpha|^2} = \frac{1}{|\alpha|}$

(d) 相干态是具有最小不确定性的量子态

一般情况：

$[\hat{x}, \hat{p}] = i\hbar$,

不确定关系（海森伯）

若 $[\hat{F}, \hat{G}] = i\hat{C}$，则

$$\overline{(\Delta\hat{F})^2} \cdot \overline{(\Delta\hat{G})^2} \geq \frac{\overline{\hat{C}^2}}{4}$$

$$\overline{(\Delta\hat{x})^2} \equiv \langle\hat{x}^2\rangle - \langle\hat{x}\rangle^2 \quad \text{(均方偏差)}$$

$$\overline{(\Delta\hat{p})^2} \equiv \langle\hat{p}^2\rangle - \langle\hat{p}\rangle^2$$

则 $$\overline{(\Delta\hat{x})^2} \cdot \overline{(\Delta\hat{p})^2} \geq \frac{\hbar^2}{4} \quad \text{(不确定关系)}$$

当 $\overline{(\Delta\hat{x})^2} \cdot \overline{(\Delta\hat{p})^2} = \frac{\hbar^2}{4}$ 时，对应量子态称为具有最小不确定性的量子态.

· $\langle \alpha t | \hat{x} | \alpha t \rangle$

$= \langle \alpha t | \sqrt{\frac{\hbar}{2m\omega}} (\hat{a}^\dagger + \hat{a}) | \alpha t \rangle$

$= \sqrt{\frac{\hbar}{2m\omega}} \left((\alpha e^{-i\omega t})^* + (\alpha e^{-i\omega t}) \right)$

$= \sqrt{\frac{\hbar}{2m\omega}} \, 2 \, \mathrm{Re}[\alpha e^{-i\omega t}], \quad (= \bar{x}_\alpha(t))$

· $\langle \alpha t | \hat{x}^2 | \alpha t \rangle = \frac{\hbar}{2m\omega} \left(1 + 4 \left(\mathrm{Re}(\alpha e^{-i\omega t}) \right)^2 \right)$,

· $\langle \alpha t | \hat{p} | \alpha t \rangle = \sqrt{\frac{m\omega\hbar}{2}} \, 2 \, \mathrm{Im}(\alpha e^{-i\omega t})$, $(= \bar{p}_\alpha(t))$

· $\langle \alpha t | \hat{p}^2 | \alpha t \rangle = \frac{m\omega\hbar}{2} \left[1 + 4 \left(\mathrm{Im}(\alpha e^{-i\omega t}) \right)^2 \right]$

$\Rightarrow \quad \overline{(\Delta\hat{x})^2} = \frac{\hbar}{2m\omega} \quad ; \quad \overline{(\Delta\hat{p})^2} = \frac{m\omega\hbar}{2}$

$\Rightarrow \quad \overline{(\Delta\hat{x})^2} \cdot \overline{(\Delta\hat{p})^2} = \frac{\hbar^2}{4}$.

. (e) 相干态是最接近经典谐振子的量子态.

① 经典谐振子运动方程:

$$m\ddot{x} + \omega^2 x = 0 .$$

解: $$\underline{x(t) = I_0 \cos(\omega t - \phi)}$$

I_0: 振幅; ϕ: 相位.

· $x(t) \longrightarrow \langle \alpha, t | \hat{x} | \alpha, t \rangle \equiv \overline{\hat{x}_\alpha(t)}$

(量子力学)

$$\underline{\overline{\hat{x}_\alpha(t)} = \langle \alpha, t | \hat{x} | \alpha, t \rangle} = \sqrt{\frac{\hbar}{2m\omega}} 2\mathrm{Re}\left(\alpha e^{-i\omega t}\right)$$

$$= \sqrt{\frac{2\hbar}{m\omega}} |\alpha| \cos(\omega t - \phi) ,$$

令 $\alpha = |\alpha| e^{i\phi}$,

$$\Rightarrow \underline{\overline{\hat{x}_\alpha(t)} = I_\alpha \cos(\omega t - \phi)}$$

其中, 振幅 $\underline{I_\alpha \equiv \sqrt{\frac{2\hbar}{m\omega}} |\alpha|}$.

② 系统总能量

经典谐振子总能量：

$$E_c = \frac{m\omega^2}{2} I_0^2$$

- $E \longrightarrow \langle \alpha t | \hat{H} | \alpha t \rangle \equiv \overline{\hat{H}(t)}$

（量子力学）

- $\langle \alpha, t | \hat{H} | \alpha, t \rangle = \langle \alpha t | (\hat{a}^\dagger \hat{a} + \frac{1}{2}) \hbar\omega | \alpha t \rangle$

$$= |\alpha|^2 \hbar\omega + \frac{1}{2}\hbar\omega$$

$$= \frac{m\omega^2}{2} I_\alpha^2 + \underbrace{\frac{1}{2}\hbar\omega}_{\text{零点振动能}},$$

$$\overline{\hat{H}(t)} \Rightarrow \frac{m\omega^2}{2} I_\alpha^2$$